**Year: 2026-2027** **Sem V-NEP**

**Course: ASD&D Lab**

**Experiment 05**

**Date of Performance: 22/07/2026** **Date of Submission: 22/07/2026**

**Class: T1**

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**Experiment 04**

**Aim:** To understand continuous integration, Jenkins installation

This experiment maps to LO1,LO3:

LO1:Apply Agile methodologies and use JIRA for sprint planning, task tracking and integrating DevOp workflows

LO3:Build and Manage automated CI/CD Pipeline for application development, testing, packaging, and deployment.

Theory:

**Jenkins**

Jenkins is a free and open-source automation server widely used to implement **Continuous Integration (CI)** and **Continuous Delivery/Deployment (CD)** in software development. It automates repetitive tasks such as compiling source code, executing automated tests, generating reports, packaging applications, and deploying software to different environments. By automating these processes, Jenkins helps improve software quality, reduces manual effort, and accelerates the development lifecycle.

Jenkins integrates with a wide range of development tools, including **Git, Maven, Gradle, JUnit**, Docker, Kubernetes, and various cloud platforms through its extensive plugin library. It is built using **Java** and runs on the **Java Virtual Machine (JVM)**. Jenkins also provides a

web-based interface that allows developers to configure jobs, monitor builds, manage users, install plugins, and create automated pipelines.

**Jenkins Pipeline**

A **Jenkins Pipeline** is a sequence of automated tasks that defines the complete software delivery workflow. The pipeline is written as code in a file called a **Jenkinsfile**, allowing the entire build and deployment process to be version-controlled along with the application source code. This approach improves consistency, maintainability, and collaboration among development teams.

A typical Jenkins pipeline consists of the following stages:

- **Build** – Compiles the application's source code and creates executable files or packages.
- **Test** – Executes automated tests to verify the application's functionality, quality, and reliability.
- **Deploy** – Deploys the successfully tested application to a testing, staging, or production environment.

**Advantages of Jenkins Pipeline**

- Automates the entire software development and deployment process.
- Minimizes human intervention and reduces manual errors.
- Speeds up code integration, testing, and software delivery.
- Provides clear visibility into each stage of the build process.
- Supports Continuous Integration (CI) and Continuous Deployment (CD) practices.
- Stores pipeline configurations as code, making them easy to maintain, share, and version-control.

**Types of Jenkins Pipelines**

**1. Declarative Pipeline**

The **Declarative Pipeline** is the recommended and most commonly used pipeline syntax in Jenkins. It follows a structured format using the pipeline {} block, making the pipeline easier to understand, write, and maintain. Due to its simple and organized syntax, it is especially suitable for beginners and teams that prefer a standardized pipeline structure.

**Features:**

- Uses the pipeline {} syntax.
- Provides a simple and well-defined structure.
- Easier to read, write, and maintain.
- Ideal for beginners and standard CI/CD workflows.

**Example:**

```
        echo 'Building application'
    }
}
        Jenkins plays an important role in DevOps by automating the software development lifecycle,
        improving efficiency, reducing errors, and enabling faster and more reliable software
        delivery.
    }
}
```

- Scripted Pipeline
  - Written using Groovy scripting.
  - Provides more flexibility and control.
  - Suitable for complex workflows.
  - Requires more programming knowledge.

  Example:

Implementation

1. Check if JDK (Java Development Kit) is installed.

```
C:\Users\15L>javac -versio
javac 21.0.2
```

2. Download Jenkins WAR file from official website or packet manager

# jenkins.war

permalink to the latest

| | |
|---|---|
| 2.575 | Released: Jul 28, 2026<br>SHA-1: 781df4433fa03537c429df8b909d2f3ebeb0d8d3<br>SHA-256: 183c18356b76b6680c7d2f689258acc8899aff6c7573d6a0e3ae0852f85b2bfc |
| 2.574 | Released: Jul 21, 2026<br>SHA-1: 143d74dfc83f5f124c5b6a5ac262b1788c1ba483<br>SHA-256: e924a55dff7e96efed4844df55c6433b8dc03c7e639b45f8289e6df8b893fca2 |
| 2.573 | Released: Jul 14, 2026<br>SHA-1: bde1bd390e74d55d202f34047c22f303c631c8b5<br>SHA-256: 8e7235833e8de1da0089569abc8754cf561bcd1bc31e98ca09b7aef1949f3109 |
| 2.572 | Released: Jul 7, 2026<br>SHA-1: 680ecd36004256c1b8f948d9fbff6e80bbd92fe6<br>SHA-256: 0c3ea851b126ee86a921822a3b996354975 3e629a52d7a4e3b3378e3d5e23f28 |
| 2.571 | Released: Jun 30, 2026<br>SHA-1: f782160fcf62147d835edc00d2c3dbdadb426b23<br>SHA-256: 3b9100b27639b1465c43cabe73acda51c2e78424fb8114fd499cd970f667d02e |
| 2.570 | Released: Jun 23, 2026<br>SHA-1: 0f423bd3baefaf8f33ffe62d14efd633674a61e1<br>SHA-256: 64404d9b406dba0f627679139374 8dddff6044c2d8d82afaef3a86d9667eeceb |
| 2.569 | Released: Jun 16, 2026<br>SHA-1: adb1898475c3d1f011e1d7fd0dc58db54cc55181<br>SHA-256: 9aa4df8f1d6515931bf25b07b886f2fc706a94c955dc1cbb39c21b8fa50f27c4e |
| 2.568.1 | Released: Jul 9, 2026<br>SHA-1: 89be8e837d5fb39d7350fccce8c58711797c1499<br>SHA-256: 58f2d4 3965fbef77 0 0629fbe158d51bf138ffd577cadbc86bd6367e8ad0beb83 |

3. Launch command prompt on your system and give the following command

```
C:\Users\15L>java -jar "C:\Program Files\Jenkins\jenkins.war"
Running from: C:\Program Files\Jenkins\Jenkins.war
webroot: C:\Users\15L\.jenkins\war
2026-08-03 10:04:42.127+0000 [id=1]     INFO    winstone.Logger#logInternal: Beginning extraction from war file
2026-08-03 10:04:43.210+0000 [id=1]     WARNING o.e.j.e9.nested.ContextHandler#setContextPath: Empty contextPath
2026-08-03 10:04:43.249+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStart: jetty-12.1.5; built: 2025-12-03T22:18:24.732Z; git: 4905d09d7b6801e792d7b73946cf7b66e4be25d6; jvm 21.0.2+13-LTS-58
2026-08-03 10:04:43.257+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStop: Stopped oejs.Server@54eb2b70{STOPPING}[12.1.5,sto=0]
2026-08-03 10:04:43.257+0000 [id=1]     INFO    winstone.Logger#logInternal: Jetty shutdown successfully
java.io.IOException: Failed to start Jetty
        at Jenkins Main ClassLoader//winstone.Launcher.<init>(Launcher.java:197)
        at Jenkins Main ClassLoader//winstone.Launcher.main(Launcher.java:493)
        at java.base/jdk.internal.reflect.DirectMethodHandleAccessor.invoke(DirectMethodHandleAccessor.java:103)
        at java.base/java.lang.reflect.Method.invoke(Method.java:580)
        at executable_Main.main(Main.java:132)
Caused by: java.io.IOException: Failed to bind to 0.0.0.0/0.0.0.0:8080
        at Jenkins Main ClassLoader//org.eclipse.jetty.server.ServerConnector.openAcceptChannel(ServerConnector.java:349)
        at Jenkins Main ClassLoader//org.eclipse.jetty.server.ServerConnector.open(ServerConnector.java:313)
        at Jenkins Main ClassLoader//org.eclipse.jetty.server.Server.lambda$doStart$0(Server.java:617)
        at java.base/java.util.stream.ForEachOps$ForEachOp$OfRef.accept(ForEachOps.java:184)
        at java.base/java.util.stream.ReferencePipeline$151.accept(ReferencePipeline.java:197)
        at java.base/java.util.stream.ReferencePipeline$251.accept(ReferencePipeline.java:179)
        at java.base/java.util.Spliterators$ArraySpliterator.forEachRemaining(Spliterators.java:1024)
        at java.base/java.util.stream.AbstractPipeline.copyInto(AbstractPipeline.java:509)
        at java.base/java.util.stream.AbstractPipeline.wrapAndCopyInto(AbstractPipeline.java:499)
        at java.base/java.util.stream.ForEachOps$ForEachOp.evaluateSequential(ForEachOps.java:151)
        at java.base/java.util.stream.ForEachOps$ForEachOp$OfRef.evaluateSequential(ForEachOps.java:174)
        at java.base/java.util.stream.AbstractPipeline.evaluate(AbstractPipeline.java:234)
        at java.base/java.util.stream.ReferencePipeline.forEach(ReferencePipeline.java:596)
        at Jenkins Main ClassLoader//org.eclipse.jetty.server.Server.doStart(Server.java:613)
        at Jenkins Main ClassLoader//org.eclipse.jetty.util.component.AbstractLifeCycle.start(AbstractLifeCycle.java:92)
        at Jenkins Main ClassLoader//winstone.Launcher.<init>(Launcher.java:193)
        ... 4 more
Caused by: java.net.BindException: Address already in use: bind
        at java.base/sun.nio.ch.Net.bind0(Native Method)
        at java.base/sun.nio.ch.Net.bind(Net.java:565)
        at java.base/sun.nio.ch.ServerSocketChannelImpl.netBind(ServerSocketChannelImpl.java:344)
        at java.base/sun.nio.ch.ServerSocketChannelImpl.bind(ServerSocketChannelImpl.java:301)
        at Jenkins Main ClassLoader//org.eclipse.jetty.server.ServerConnector.openAcceptChannel(ServerConnector.java:344)
        ... 19 more
2026-08-03 10:04:43.258+0000 [id=1]     SEVERE  winstone.Logger#logInternal: Container startup failed
java.net.BindException: Address already in use: bind
        at java.base/sun.nio.ch.Net.bind0(Native Method)
        at java.base/sun.nio.ch.Net.bind(Net.java:565)
        at java.base/sun.nio.ch.ServerSocketChannelImpl.netBind(ServerSocketChannelImpl.java:344)
        at java.base/sun.nio.ch.ServerSocketChannelImpl.bind(ServerSocketChannelImpl.java:301)
        at Jenkins Main ClassLoader//org.eclipse.jetty.server.ServerConnector.openAcceptChannel(ServerConnector.java:344)
Caused: java.io.IOException: Failed to bind to 0.0.0.0/0.0.0.0:8080
        at Jenkins Main ClassLoader//org.eclipse.jetty.server.ServerConnector.openAcceptChannel(ServerConnector.java:349)
```

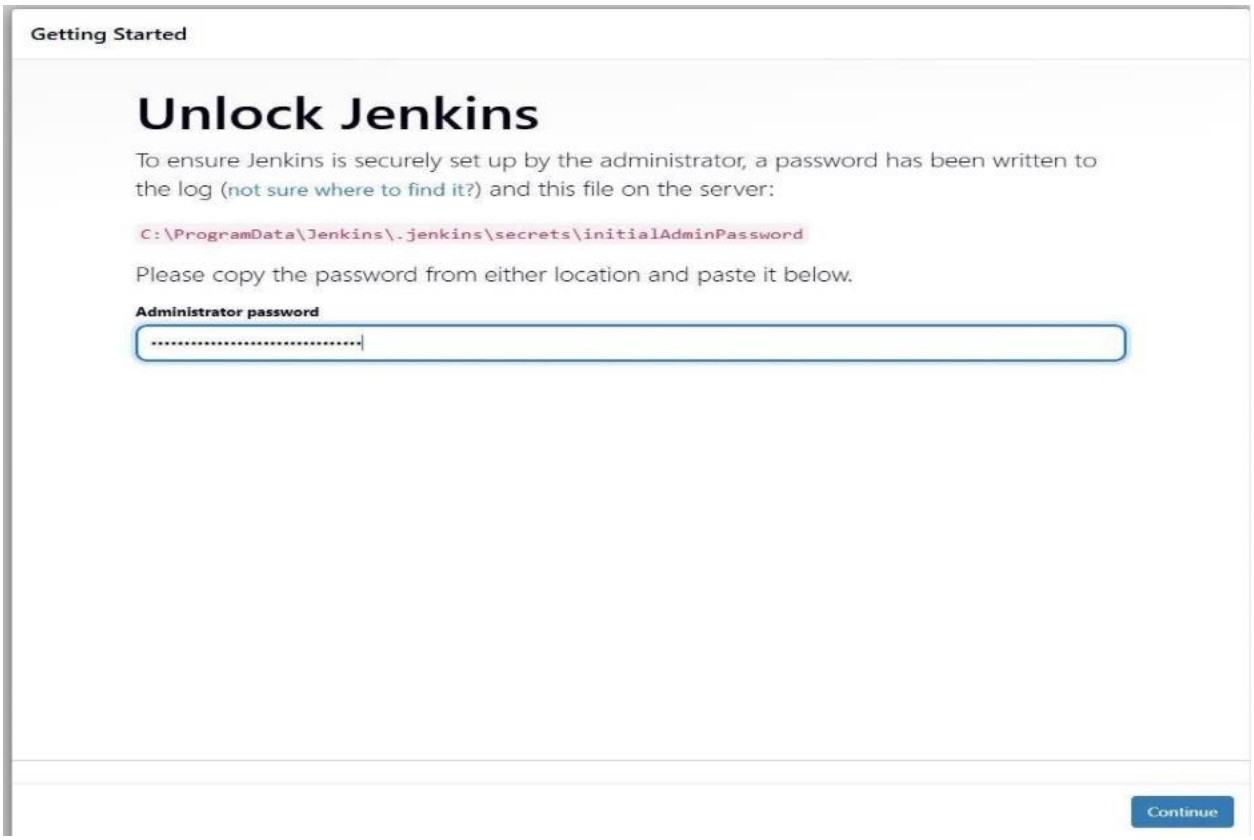
4. Navigate to the folder

```
        at executable.Main.main(Main.java:332)

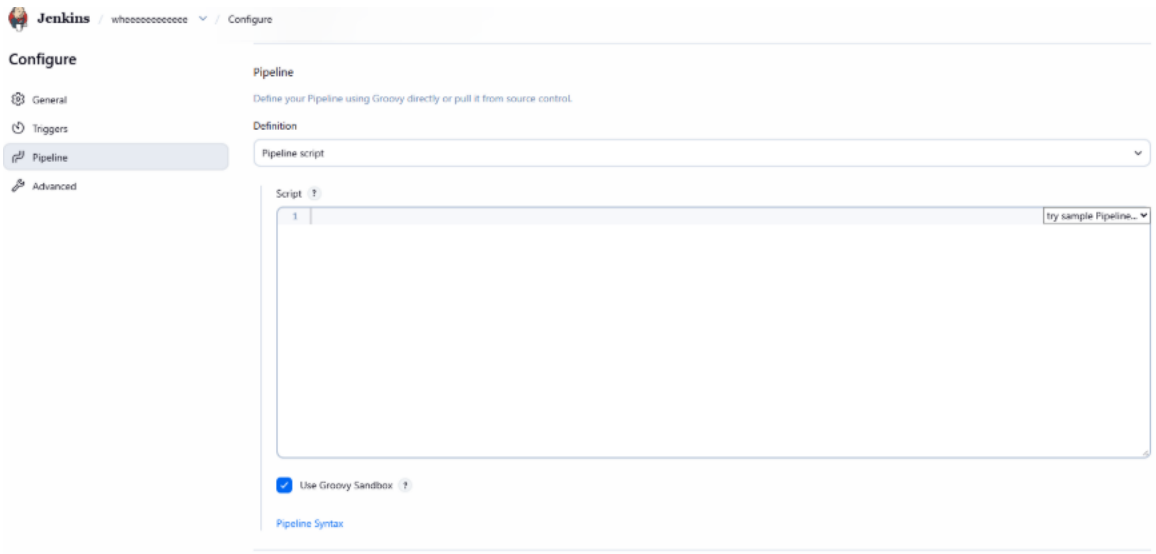
C:\Users\15L>cd "C:\Program Files\Jenkins"

C:\Program Files\Jenkins>java -jar jenkins.war
Running from: C:\Program Files\Jenkins\Jenkins.war
webroot: C:\Users\15L\.jenkins\war
2026-08-03 10:07:11.878+0000 [id=1]     INFO    winstone.Logger#logInternal: Beginning extr
2026-08-03 10:07:11.941+0000 [id=1]     WARNING o.e.j.ee9.nested.ContextHandler#setContextP
2026-08-03 10:07:12.020+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStart: je
2026-08-03 10:07:12.030+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStop: Sto
2026-08-03 10:07:12.031+0000 [id=1]     INFO    winstone.Logger#logInternal: Jetty shutdown
java.io.IOException: Failed to start Jetty
        at Jenkins Main ClassLoader//winstone.Launcher.<init>(Launcher.java:197)
        at Jenkins Main ClassLoader//winstone.Launcher.main(Launcher.java:493)
        at java.base/jdk.internal.reflect.DirectMethodHandleAccessor.invoke(DirectMethodHan
        at java.base/java.lang.reflect.Method.invoke(Method.java:580)
        at executable.Main.main(Main.java:332)
```

5. Start Jenkins: Run the command C:\Program Files\Jenkins>java -jar jenkins.war to start Jenkins server.
6. Access Web UI: open localhost 8090



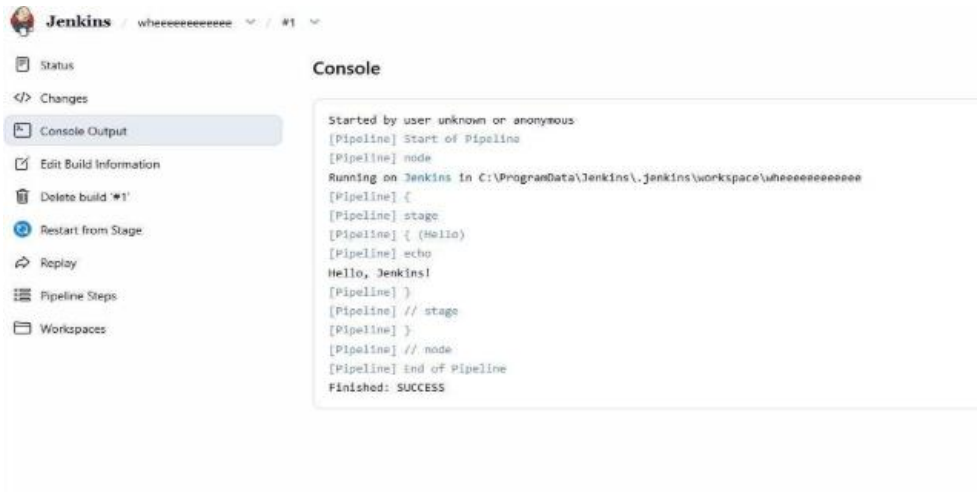

7. Open jenkins and pipeline




Code:

```
pipeline {
    agent any
    stages {
        stage('Hello') {
            steps {
                echo 'Hello, Jenkins!'
            }
        }
    }
}
```

Output:




Conclusion:

Jenkins Pipeline automates the software development process by integrating stages like build, test, and deployment to support Continuous Integration and Continuous Delivery. In this experiment, Jenkins was successfully installed and configured by verifying the JDK, launching Jenkins using the jenkins.war file, and accessing the setup page through a browser. A Declarative Pipeline was created and executed, demonstrating how Jenkins automates tasks and improves the efficiency and reliability of the CI/CD process.